

3.2 String slicing ^

Settings

Beta features

index

An index is an integer matching a specific position in a string's sequence.

Slice notation

Slice notation has the form `my_str[start:end]`, which creates a new string that contains the characters of `my_str` from indices start to end - 1.

Figure 3.2.1: String slicing.

```
url = "http://en.wikipedia.org/wiki/Turing"
domain = url[7:23] # Read "en.wikipedia.org" from url
print(domain)
```

en.

PARTICIPATION ACTIVITY

3.2.1: Slicing.

Start ☐ 2x speed

	0	1	2	3	4	5	6	7	8	9	10
my_str =	D	O		N	O	T		L	I	E	!

my_str[0:2] : "DO"

my_str[0:6] : "DO NOT"

my_str[7:10] : "LIE"

Captions ^

1. my_str[0:2] returns a substring of my_str starting at index 0 up to, but not including, index 2.
2. my_str[0:6] returns a substring of my_str starting at index 0 up to, but not including, index 6.
3. my_str[7:10] returns a substring of my_str starting at index 7 up to, but not including, index 10.

Figure 3.2.2: A slice creates a new object.

```
my_str = "The cat jumped the brown cow"
animal = my_str[4:7]
print(f"The animal is a {animal}")

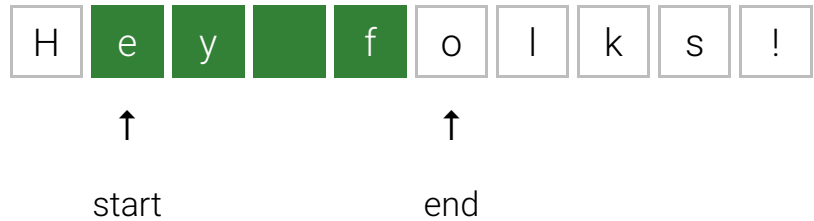
my_str = "The fox jumped the brown llama"
print(f"The animal is still a {animal}") # animal variable remains unchanged.
```

PARTICIPATION ACTIVITY

3.2.3: String slicing tool.

```
string_var = "Hey folks!"  
print(string_var[1:5])
```

Output: "ey f"



Try 3.2.1: Slicing example: omitting start, end indices.

Run the program below.



▶ Run

```
1  usr_text = input("Enter a string: ")
2  print()
3
4  first_half = usr_text[:len(usr_text) // 2]
5  last_half = usr_text[len(usr_text) // 2:]
6
7  print(f"The first half of the string is \"{first_half}\")
8  print(f"The second half of the string is \"{last_half}\")
```

CONSOLE

Table 3.2.1: Common slicing operations.

A list of common slicing operations a programmer might use.
Assume the value of my_str is "http://en.wikipedia.org/wiki/Nasa/"

Syntax	Result	Description
my_str[10:19]	wikipedia	Returns the characters in indices 10 to 19.
my_str[10:-5]	wikipedia.org/wiki/	Returns the characters in indices 10 to the 5th from the end.
my_str[8:]	n.wikipedia.org/wiki/Nasa/	Returns all characters from index 8 to the end.
my_str[:23]	http://en.wikipedia.org	Returns every character up to index 23.
my_str[:-1]	http://en.wikipedia.org/wiki/Nasa	Returns all but the last character.

stride

The stride determines how much to increment the index after reaching the current index.

Figure 3.2.3: Slice stride.

```
numbers = "0123456789"

print(f"All numbers: {numbers[::]}")
print(f"Every even number: {numbers[::2]}")
print(f"Every third number between 1 and 8: {numbers[1:9:3]}")
```

All numbers: 0123456789
Every even number: 02468
Every third number between 1 and 8: 147

CHALLENGE
ACTIVITY

3.2.1: Enter the output of the sliced string.

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